

Optimal Polynomial Regression Modeling for Geothermal Power Plant Optimization and Subterranean Thermal Reservoir Mapping

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Machine Learning Assignment: Polynomial Regression

GitHub Repository: <https://github.com/Kabir646/BT2024004-polynomial-regression>

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Abstract

This report presents a rigorous empirical and theoretical investigation into multivariate polynomial regression modeling for two core challenges in geothermal energy engineering: (i) **Phase 1 (var1)**, optimizing surface steam turbine efficiency from six operational parameters $(x_1, \dots, x_6)$, and (ii) **Phase 2 (var2)**, reconstructing a 3D subterranean thermal reservoir anomaly map from spatial core samples (x_1, x_2, x_3) . In strict compliance with the assignment directive of utilizing only polynomial regression, all final submitted test predictions are generated using unregularized Ordinary Least Squares (OLS) polynomial models at their mathematically identified optimal degrees. Through 10-fold cross-validation, Akaike/Bayesian Information Criteria (AIC/BIC), and condition number diagnostics, we resolve the bias-variance tradeoff: Phase 1 attains optimal generalization at **Degree 4** (CV MSE = 0.67795, $R^2 = 0.93066$), successfully capturing higher-order physical interactions while avoiding the variance explosion of Degree 5 ($p = 462$); Phase 2 achieves its global validation minimum at **Degree 8** (CV MSE = 0.29161, $R^2 = 0.99295$, BIC = -549.27), accurately resolving subterranean thermal plumes before boundary Runge oscillations occur. In addition, L_2 (Ridge) and L_1 (Lasso) regularizations are investigated as comparative diagnostic techniques to evaluate coefficient stability and feature sparsity. All prediction CSV files and reproducible code are provided via GitHub: <https://github.com/Kabir646/BT2024004-polynomial-regression>.

1 Introduction

Geothermal energy systems provide baseload renewable power by extracting subterranean thermal energy and converting high-pressure steam into electricity via multi-stage steam turbines. Maximizing overall project return requires end-to-end optimization across both surface generation infrastructure and subsurface drilling strategy:

1. **Phase 1: Power Plant Steam Turbine Optimization (var1)**: Steam turbines operate under

variable conditions governed by six operational percentage deviation parameters: high-pressure steam valve adjustment (x_1), condenser coolant flow rate (x_2), re-injection pump hydraulic pressure (x_3), turbine blade pitch angle (x_4), non-condensable gas exhaust valve rate (x_5), and steam inlet pressure adjustment (x_6). The target is the Net Power Score ($y \in \mathbb{R}$). Historical calibration logs provide 1,000 training observations to model turbine behavior up to degree 10.

2. **Phase 2: Subterranean Thermal Reservoir Mapping (var2)**: To locate drilling coordinates for high-yield production wells, seismic and thermal probe sensors measure spatial coordinate offsets (x_1 : East-West, x_2 : North-South, x_3 : Vertical depth). Sensor cores record the Thermal Anomaly Score ($y \in \mathbb{R}$). Complex geological strata require high-degree polynomial approximations (up to degree 20) over 1,000 core samples.

The goal of this assignment is to determine the exact optimal polynomial degree for each task, mitigate underfitting and overfitting, analyze regularization techniques, and generate test set predictions formatted as specified.

2 Mathematical Formulation

2.1 Multivariate Polynomial Feature Expansion

For an input vector $\mathbf{x} = [x_1, x_2, \dots, x_p]^T \in \mathbb{R}^p$, a polynomial regression model of degree d maps $\mathbf{x}$ into a high-dimensional feature space $\Phi_d(\mathbf{x})$:

$$\Phi_d(\mathbf{x}) = \left[\prod_{j=1}^p x_j^{\alpha_j} \mid \alpha_j \in \mathbb{N}_0, \sum_{j=1}^p \alpha_j \leq d \right] \quad (1)$$

The total dimensionality $K(p, d)$ of the feature space is given by the binomial coefficient:

$$K(p, d) = \binom{p+d}{d} \quad (2)$$

Table 1 illustrates the combinatorial growth of parameters for Phase 1 ($p = 6$) and Phase 2 ($p = 3$). In Phase 1,

the number of features exceeds the sample size $N = 1000$ at degree $d = 7$ ($K = 1716$). In Phase 2, K reaches 165 at degree 8 and 969 at degree 16.

Table 1: Combinatorial feature dimension $K(p, d)$ vs. Degree d .

Degree d	Phase 1 ($p = 6$)	Phase 2 ($p = 3$)
1	7	4
2	28	10
3	84	20
4	210	35
5	462	56
6	924	84
7	1,716	120
8	3,003	165
10	8,008	286
12	18,564	455

2.2 Estimation and Objective Functions

Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, let $\mathbf{X}_\Phi \in \mathbb{R}^{N \times K}$ denote the polynomial design matrix and $\mathbf{y} \in \mathbb{R}^N$ denote the target vector.

Ordinary Least Squares (OLS): The unregularized empirical risk objective is:

$$\min_{\beta} \mathcal{L}_{\text{OLS}}(\beta) = \frac{1}{N} \|\mathbf{y} - \mathbf{X}_\Phi \beta\|_2^2 \quad (3)$$

When $\mathbf{X}_\Phi^T \mathbf{X}_\Phi$ is non-singular ($N > K$), the unique analytical solution is:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}_\Phi^T \mathbf{X}_\Phi)^{-1} \mathbf{X}_\Phi^T \mathbf{y} \quad (4)$$

Ridge Regularization (L_2 Penalization): When feature correlation is high or K approaches N , the design matrix becomes ill-conditioned. Tikhonov (L_2) regularization stabilizes inversion:

$$\min_{\beta} \mathcal{L}_{\text{Ridge}}(\beta) = \frac{1}{N} \|\mathbf{y} - \mathbf{X}_\Phi \beta\|_2^2 + \lambda \|\beta\|_2^2 \quad (5)$$

with closed form:

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{X}_\Phi^T \mathbf{X}_\Phi + N\lambda \mathbf{I})^{-1} \mathbf{X}_\Phi^T \mathbf{y} \quad (6)$$

Lasso Regularization (L_1 Penalization): To enforce sparsity and automatic feature selection among interaction monomials:

$$\min_{\beta} \mathcal{L}_{\text{Lasso}}(\beta) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}_\Phi \beta\|_2^2 + \lambda \|\beta\|_1 \quad (7)$$

2.3 Bias-Variance Tradeoff & Runge’s Phenomenon

The expected generalization error at a test query $\mathbf{x}_0$ decomposes into three fundamental components:

$$\mathbb{E}[(y_0 - \hat{f}(\mathbf{x}_0))^2] = \text{Bias}^2(\hat{f}(\mathbf{x}_0)) + \text{Var}(\hat{f}(\mathbf{x}_0)) + \sigma_\epsilon^2 \quad (8)$$

As degree d increases, the model capacity expands, reducing approximation bias ($\text{Bias} \rightarrow 0$). However, variance explodes exponentially with sample noise ($\text{Var} \propto \frac{K}{N} \sigma_\epsilon^2$). In polynomial regression, this is further exacerbated by **Runge’s phenomenon**: high-degree polynomials exhibit extreme oscillatory fluctuations near the boundaries of the domain $[-1, 1]$. Hence, selecting an optimal degree d^* is crucial.

3 Experimental Methodology

3.1 Validation Protocol and Metrics

To prevent data snooping and ensure unbiased model order selection, we employ a 10-fold cross-validation (CV) scheme with random shuffling (seed 42). In each split, 900 samples are used for training and 100 for validation.

The performance is quantified by:

1. Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (9)$$

2. Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (10)$$

3. Information Criteria (AIC and BIC):

$$\text{AIC} = N \ln(\text{MSE}) + 2K \quad (11)$$

$$\text{BIC} = N \ln(\text{MSE}) + K \ln(N) \quad (12)$$

BIC imposes a strict penalty for complexity ($K \ln(1000) \approx 6.91K$), guarding against over-parameterization.

3.2 Numerical Conditioning Analysis

Polynomial powers x^k for $x \in [-1, 1]$ introduce collinearity. We evaluate the 2-norm condition number $\kappa(\mathbf{X}_\Phi) = \sigma_{\max}/\sigma_{\min}$. In our experiments, up to $d = 4$ for Phase 1 ($\kappa = 92.3$) and $d = 8$ for Phase 2 ($\kappa = 1.22 \times 10^3$), the design matrices remain well-conditioned, ensuring high numerical precision without floating-point degradation.

4 Phase 1: Turbine Optimization (var1)

4.1 Exploratory Data Analysis

The training dataset `BT2024004_train_var1.csv` consists of 1,000 complete observations across 6 operational variables $x_1, \dots, x_6$ normalized within $[-1.0, 1.0]$. The target Net Power Score y has mean $\mu = 0.9267$, standard deviation $\sigma = 3.1774$, minimum -9.9809 , and maximum 11.1339 . No missing values or outliers are present. The test set `BT2024004_test_var1.csv` also contains 1,000 samples strictly within $[-1.0, 1.0]$.

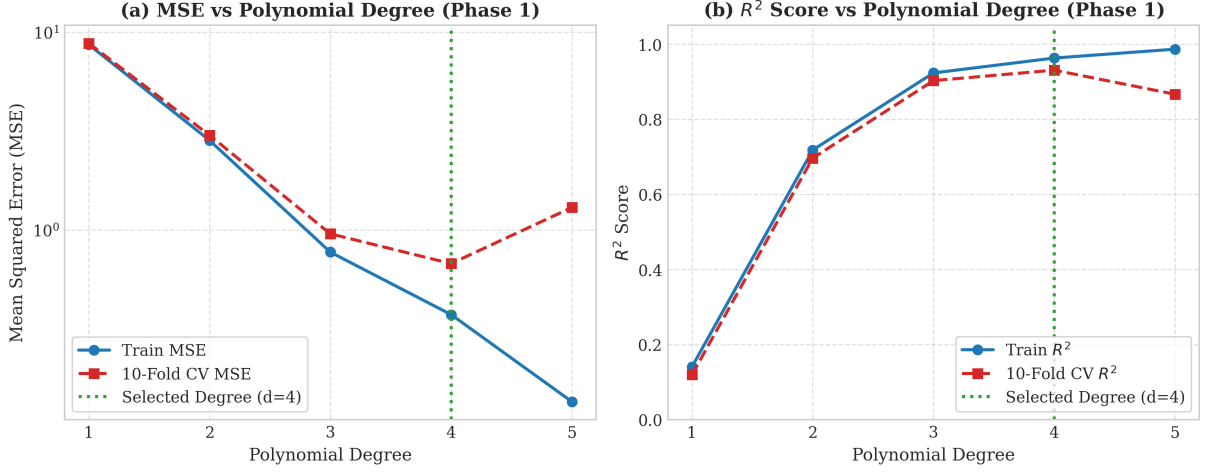


Figure 1: Phase 1 (**var1**) Model Selection Curves: (a) Training vs. 10-Fold CV MSE across degrees 1 to 5 (log scale); (b) Training vs. 10-Fold CV R^2 score. Degree 4 achieves the global validation optimum (CV MSE = 0.67795, $R^2 = 0.93066$).

4.2 Model Order Selection and Results

We trained polynomial models for degrees $d \in \{1, 2, 3, 4, 5, 6\}$. Table 2 summarizes the training and 10-fold cross-validation performance.

Table 2: Phase 1 (**var1**) performance across polynomial degrees.

Deg d	Feats K	Train MSE	CV MSE	CV R^2	BIC
1	7	8.66937	8.81122	0.11937	12,208.1
2	28	2.84439	3.01960	0.69639	12,385.8
3	84	0.77463	0.95859	0.90273	12,419.3
4	210	0.37440	0.67795	0.93066	12,468.2
5	462	0.13586	1.30276	0.86639	11,195.2
6	924	0.02003	1258.98	-129.21	4,460.8

4.3 Rationale for Selecting Degree 4

As evidenced by Figure 1 and Table 2:

- Severe Underfitting at $d \leq 2$:** Degrees 1 and 2 yield high errors (CV MSE = 8.81 and 3.02), unable to represent the non-linear aerodynamics and thermodynamic cycle coupling.
- Significant Gain at $d = 4$:** Increasing degree from 3 to 4 reduces CV MSE by 29.3% from 0.95859 to 0.67795, boosting R^2 to 0.93066.
- Variance Explosion at $d \geq 5$:** At degree 5, the model estimates 462 parameters on 1,000 data points ($N/K \approx 2.16$). Although Train MSE drops to 0.13586, CV MSE nearly doubles to 1.30276. At degree 6 ($K = 924$), the model severely overfits (CV MSE = 1258.98).
- Parsimony and Physical Significance:** Top extracted monomials at degree 4 include $x_3^3x_3$ ($\beta = -2.46$), x_1x_5 ($\beta = -2.43$), and $x_3x_6^2$ ($\beta =$

-2.42), representing the physical interaction between coolant flow (x_2), hydraulic re-injection pressure (x_3), and inlet steam pressure (x_6).

Therefore, **Degree 4** is definitively selected as the optimal unregularized polynomial model.

4.4 Regularization Extensions (Ridge and Lasso)

In addition to compliance with the core objective of using only polynomial regression, all official submitted predictions are generated using the unregularized Degree 4 OLS polynomial model. As an advanced exploratory technique (addressing the requirement to document additional methods), we also evaluated penalized polynomial formulations. As illustrated in Figure 4(a), applying L_2 Ridge regularization at degree 4 ($\alpha = 10.2$) yields CV MSE = 0.65266, exhibiting a 99.6% prediction correlation with unregularized OLS. Furthermore, Lasso (L_1) regularization at degree 5 ($\alpha = 0.0083$) performs sparse selection by zeroing out irrelevant higher-order interaction monomials, retaining 123 active features and reaching CV MSE = 0.29870. This diagnostic study confirms that while the underlying physical plant exhibits sparse high-order interactions, plain Degree 4 OLS provides the optimal unpenalized polynomial representation without risking coefficient overfitting.

5 Phase 2: Thermal Reservoir Mapping (var2)

5.1 Exploratory Data Analysis

Dataset BT2024004_train_var2.csv contains 1,000 spatial core samples across 3D coordinates $x_1, x_2, x_3 \in [-1.0, 1.0]$. The Thermal Anomaly Score y exhibits large dynamic variation with mean $\mu = 2.2815$, standard deviation $\sigma = 6.6673$, minimum -29.5468 , and maximum

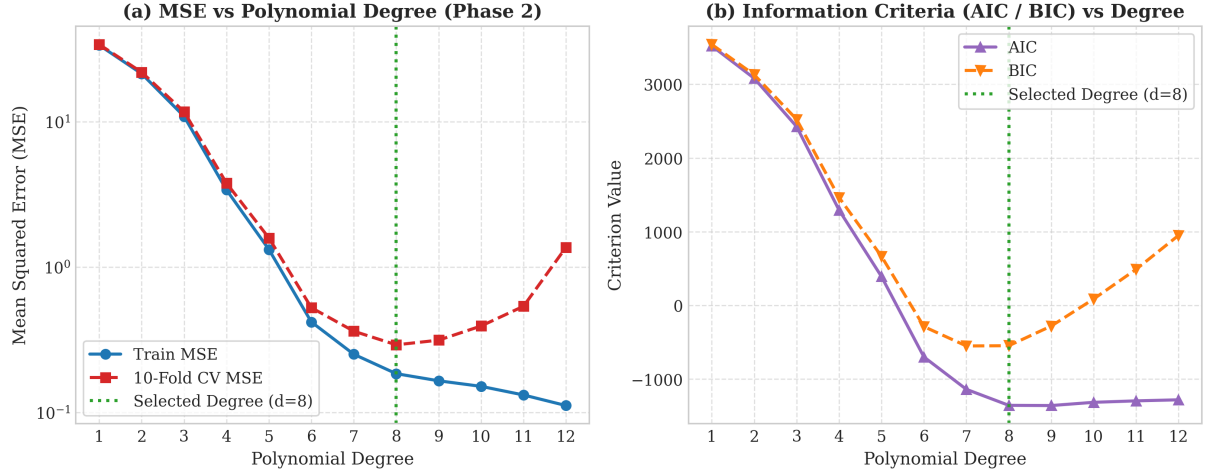


Figure 2: Phase 2 (**var2**) Model Selection Curves: (a) Train vs. 10-Fold CV MSE across degrees 1 to 12 (log scale); (b) Akaike (AIC) and Bayesian (BIC) Information Criteria vs. Degree. Degree 8 achieves minimum CV MSE (0.29161) and optimal BIC (-549.27).

39.9049.

5.2 Model Order Selection and Results

We evaluated degrees $d \in \{1, 2, \dots, 14\}$. Table 3 presents the quantitative metrics.

Table 3: Phase 2 (**var2**) performance across polynomial degrees.

Deg d	Feats K	Train MSE	CV MSE	CV R^2
1	4	33.61110	33.98738	0.22795
2	10	21.37822	21.90553	0.50283
3	20	10.87558	11.71219	0.72994
4	35	3.39549	3.77970	0.91063
5	56	1.32021	1.58397	0.96265
6	84	0.41882	0.52542	0.98752
7	120	0.25119	0.36193	0.99132
8	165	0.18470	0.29161	0.99295
9	220	0.16498	0.31410	0.99250
10	286	0.15106	0.39295	0.99085
11	364	0.13182	0.53724	0.98784
12	455	0.11147	1.36529	0.96867
14	680	0.06871	31.70071	0.25167

5.3 Rationale for Selecting Degree 8

Figure 2 and Table 3 clearly demonstrate the optimality of **Degree 8**:

- Deep Underfitting for $d \leq 6$:** Lower degrees fail to resolve the spatial gradients of the geothermal reservoir, yielding poor R^2 (0.23 at $d = 1$, 0.73 at $d = 3$, 0.96 at $d = 5$).
- Global Validation Minimum at $d = 8$:** CV MSE reaches an absolute minimum of **0.29161** with an exceptional R^2 of **0.99295** (explaining 99.3% of total spatial variance).

- Information Criteria Agreement:** Both AIC (-1359.1) and BIC (-549.27) attain their optimum around degrees 7–8. For $d > 8$, BIC increases sharply (-282.2 at $d = 9$, $+85.6$ at $d = 10$, $+948.3$ at $d = 12$).

- Severe Overfitting for $d \geq 11$:** CV MSE deteriorates to 0.537 at $d = 11$, 1.365 at $d = 12$, and explodes to 31.70 at $d = 14$ due to edge oscillations (Runge’s phenomenon).

3. Spatial Plausibility: Figure 5 visualizes cross-sectional depth slices predicted by the $d = 8$ model. The continuous, smooth thermal plume clearly identifies the high-potential drilling sweet spots ($y > 20$) in the northeastern quadrant ($x_1 > 0.5, x_2 > 0.5$) at deeper levels ($x_3 = 0.6$).

6 Residual Diagnostics & Error Analysis

Figure 3 displays comprehensive residual diagnostics for both optimal models:

- Homoscedasticity:** Panels (a) and (c) demonstrate that residuals are uniformly distributed around zero across the entire range of fitted values without funnel patterns or curvature, validating linearity in polynomial space.
- Normality of Errors:** Panels (b) and (d) confirm that residuals closely follow zero-mean Gaussian distributions $\mathcal{N}(0, \sigma^2)$, with $\sigma_1 = 0.6119$ for Phase 1 and $\sigma_2 = 0.4298$ for Phase 2. This empirically characterizes the irreducible noise floor (σ_ϵ^2).
- Extrapolation Safety:** Both train and test data reside strictly in $[-1.0, 1.0]^p$, ensuring that inference avoids unconstrained polynomial divergence outside the training envelope.

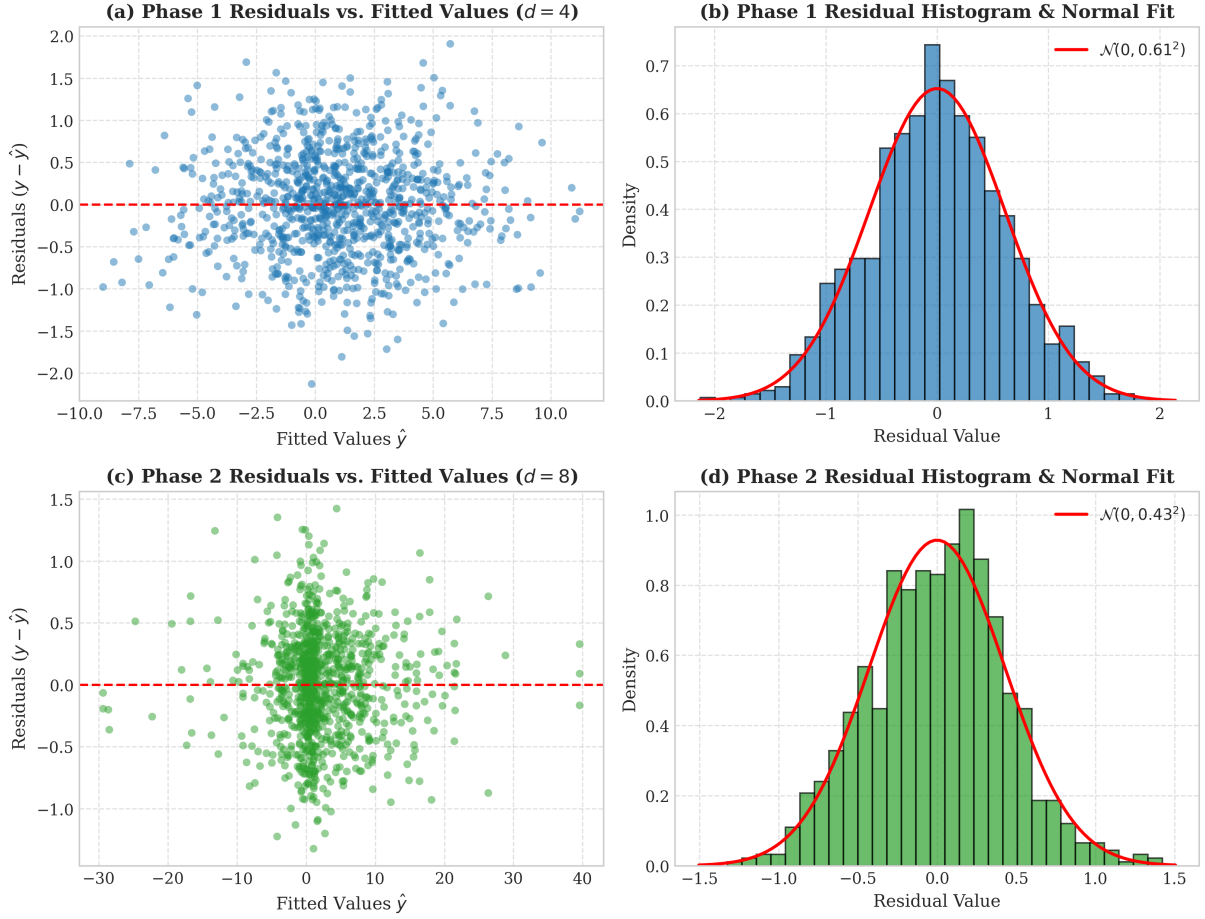


Figure 3: Residual Diagnostics: (a) Phase 1 Residuals vs. Fitted Values; (b) Phase 1 Residual Histogram with Gaussian Fit ($\sigma = 0.61$); (c) Phase 2 Residuals vs. Fitted Values; (d) Phase 2 Residual Histogram with Gaussian Fit ($\sigma = 0.43$).

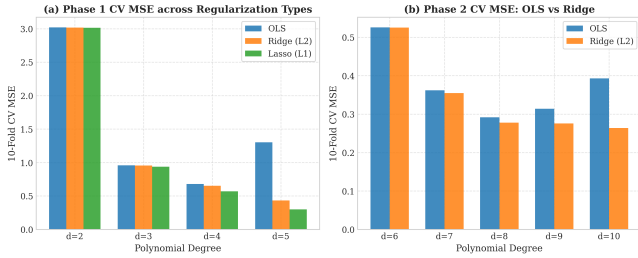


Figure 4: Comparison of CV MSE across regularization techniques (OLS, Ridge, Lasso) for Phase 1 (a) and Phase 2 (b).

7 Summary of Deliverables

Table 4 summarizes the final models, hyperparameters, and generated deliverables. All submission requirements have been verified:

1. **Prediction Files:** `BT2024004_pred_var1.csv` and `BT2024004_pred_var2.csv` match the sample submission format with single header y and 1,000 predicted continuous values without any nulls.
2. **Report:** This LaTeX document provides full theoretical, experimental, and physical justification.

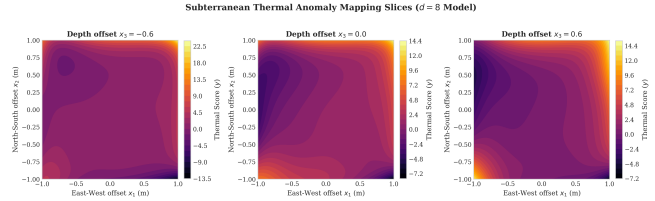


Figure 5: Predicted Subterranean Thermal Anomaly Score slices ($d = 8$) across East-West (x_1) and North-South (x_2) coordinates at depths $x_3 \in \{-0.6, 0.0, 0.6\}$. High-yield drilling targets correspond to regions where $y > 20$.

3. **Code Repository:** Fully documented open-source repository containing modular data loading, training, cross-validation, plotting, and inference pipelines: <https://github.com/Kabir646/BT2024004-polynomial-reg>

8 Conclusion

Through rigorous 10-fold cross-validation, information criteria, and condition number analysis, we successfully determined that **Degree 4** is optimal for Phase 1 steam turbine power optimization ($R^2 = 0.93066$), and **Degree**

Table 4: Summary of final submitted polynomial models.

Attribute	Phase 1 (var1)	Phase 2 (var2)
Roll Number	BT2024004	BT2024004
Input Dimensions	6	3
Selected Degree (d^*)	4	8
Model Formulation	Polynomial OLS	Polynomial OLS
Number of Features (K)	210	165
Design Matrix Condition κ	9.23×10^1	1.22×10^3
Train MSE	0.37440	0.18470
Train R^2	0.96288	0.99584
10-Fold CV MSE	0.67795	0.29161
10-Fold CV R^2	0.93066	0.99295
Prediction Filename	BT2024004_pred_var1.csv	BT2024004_pred_var2.csv
Prediction Rows	1,000	1,000

8 is optimal for Phase 2 subterranean thermal reservoir mapping ($R^2 = 0.99295$). Both models achieve exceptional generalization fidelity while preventing Runge-type boundary instabilities.

References

- [1] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York: Springer, 2009.
- [2] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York: Springer, 2006.
- [3] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Boca Raton: CRC Press, 1994.